

For questions 1-5, factor each quadratic expression using the method of your choice.

1. $5b^2 + 25b + 20$

$$5(b^2 + 5b + 4)$$

$$5(b + 1)(b + 4)$$

2. $4a^2 - 13a - 35$

$$(4a + 7)(a - 5)$$

3. $2x^2 + 5x - 33$

$$(2x + 11)(x - 3)$$

4. $3k^2 - 27k + 60$

$$(3k - 15)(k - 4)$$

$$3(k - 5)(k - 4)$$

5. $2z^2 + 13z - 70$

$$(2z - 7)(z + 10)$$

For questions 6-7, consider the trinomial $3x^2 - 2x - 5$.

6. Complete the area model.

	$3x$	<u>-5</u>
x	$3x^2$	<u>$-5x$</u>
<u>$+1$</u>	<u>$3x$</u>	-5

7. What are the two binomial factors for $3x^2 - 2x - 5$?

$(3x - 5)$ and $(x + 1)$

For questions 8-9, consider the trinomial $2m^2 + 11m - 63$.

8. Complete the area model.

	<u>$2m$</u>	<u>-7</u>
<u>m</u>	<u>$2m^2$</u>	<u>$-7m$</u>
<u>$+9$</u>	<u>$18m$</u>	<u>-63</u>

9. What are the two binomial factors for $2m^2 + 11m - 63$?

$(2m - 7)$ and $(m + 9)$

10. Factor $3x^2 + 22x + 24$ using factoring by grouping.

$(3x^2 + 18x) + (4x + 24)$
 $3x(x + 6) + 4(x + 6)$
 $(3x + 4)(x + 6)$